

Playwright TypeScript Guide

Callbacks, Promises, and Locators

1. Callback Functions in Playwright

- A callback is a function passed as an argument and executed later.
- Playwright uses callbacks in `test()`, `beforeEach()`, `page.on()`, `evaluate()`, and `waitForFunction()`.
- Callbacks are usually async functions returning Promise.

Example:

```
page.on('dialog', async (dialog) => { await dialog.accept(); });
```

2. Promises in Playwright

- A Promise represents a value that will be available in the future.
- Promise states: Pending, Fulfilled, Rejected.
- Every Playwright action returns a Promise.
- `async/await` is syntax sugar built on top of Promises.

Example:

```
await page.click('#login');  
await page.fill('#email', 'test@test.com');
```

3. Locators in Playwright

- Locators are Playwright's recommended way to interact with elements.
- They auto-wait and retry before performing actions.
- Prefer getByRole, getByText, and getByLabel over CSS or XPath.

Examples:

```
page.getByRole('button', { name: 'Login' }).click();
page.getByLabel('Email').fill('test@test.com');
page.locator('#submit').click();
```